

Analysis of Variation(ANOVA) AND PARAMETRIC TEST

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setting a working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(dplyr)
library(tidyverse)
library(ggplot2)
```

import data

```
hg <- read.csv("gsssubset.csv")
head(hg)
```

	id	sex	degree	income	marital	age	height	weight	hrswrk
1	1	MALE	BACHELOR	60967.50	DIVORCED	53	72	190	60
2	2	FEMALE	BACHELOR	60967.50	MARRIED	26	60	97	40
3	4	FEMALE	BACHELOR	10161.25	MARRIED	56	68	160	20
4	14	FEMALE	HIGH SCHOOL	17551.25	MARRIED	40	65	156	37
5	16	MALE	HIGH SCHOOL	17551.25	MARRIED	56	NA	210	6
6	19	MALE LT	HIGH SCHOOL	15703.75	MARRIED	51	68	170	NA

one way Anova

test whether there are significance differences btn group means

```
data7 <- data.frame(group = factor(c("A", "A", "A", "B", "B", "B", "C", "C", "C")),
  value = c(5:13))
head(data7)
```

	group	value
1	A	5
2	A	6
3	A	7
4	B	8
5	B	9
6	B	10

perform 1 way ANOVA

HO: there is no significance difference across the scores across the three groups

H1: there is significance scores across the three groups

```
one_anova <- aov(value ~ group, data = data7)
summary(one_anova)
```

```

      Df Sum Sq Mean Sq F value Pr(>F)
group    2     54      27      27 0.001 ***
Residuals 6       6       1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

results

- since the p-value<5%, we reject null hypothesis and conclude that exist significance btn the 3 groups

post-hoc test(turkey's hsd) : identify which group differ significantly after a significant anova results

```

turkey_results = TukeyHSD(one_anova)
print(turkey_results)

```

```

Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = value ~ group, data = data7)

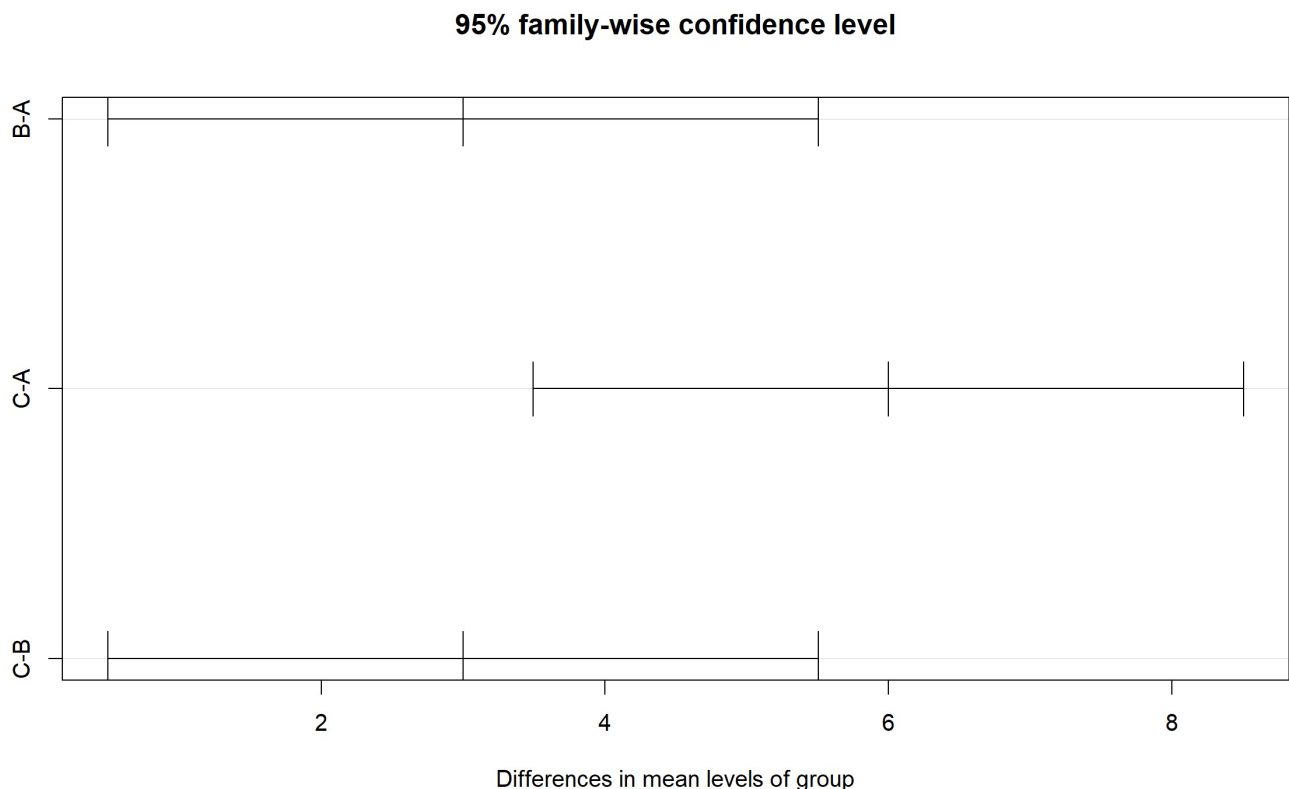
$group
      diff      lwr      upr      p adj
B-A     3 0.4947644 5.505236 0.0242291
C-A     6 3.4947644 8.505236 0.0007942
C-B     3 0.4947644 5.505236 0.0242291

```

- b-a exist

plot the results

```
plot(turkey_results)
```



2-way Anova

```

data9 <- data.frame(factor1 = factor(c("X", "X", "X", "Y", "Y", "Y")), factor2 = factor(c("P",
"Q", "R", "P", "Q", "R")), value1 = c(5:10))
head(data9)

```

	factor1	factor2	value1
1	X	P	5
2	X	Q	6
3	X	R	7
4	Y	P	8
5	Y	Q	9
6	Y	R	10

perform 2 way anova

H0: there is no significance difference on avarage income across factor 1

H1: there is no significance difference on avarage income across factor 2

H2: there is no significance difference on avarage income across the interaction

```
anova_model1 <- lm(value1 ~ factor1 * factor2, data = data9)
anova_result <- anova(anova_model1)
print(anova_result)
```

Analysis of Variance Table

Response: value1

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor1	1	13.5	13.5	NaN	NaN
factor2	2	4.0	2.0	NaN	NaN
factor1:factor2	2	0.0	0.0	NaN	NaN
Residuals	0	0.0	NaN		

using real data set

H0: there is no significance difference on avarage income across gender

H1: there is no significance difference on avarage income across marital status

H2: there is no significance difference on avarage income across the interaction

```
hg$marital <- as.factor(hg$marital)
hg$sex <- as.factor(hg$sex)
anova_model_data <- lm(income ~ sex * marital, data = hg)
anova_tway <- anova(anova_model_data)
print(anova_tway)
```

Analysis of Variance Table

Response: income

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
sex	1	8.7751e+10	8.7751e+10	84.1939	< 2.2e-16 ***
marital	4	5.8740e+10	1.4685e+10	14.0897	3.477e-11 ***
sex:marital	4	2.3753e+10	5.9382e+09	5.6975	0.0001555 ***
Residuals	984	1.0256e+12	1.0422e+09		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- the data is significance

by degree

H0: there is no significance difference on average income across degree

H1: there is no significance difference on average income across marital status

H2: there is no significance difference on average income across the interaction

```
hg$marital <- as.factor(hg$marital)
hg$degree <- as.factor(hg$degree)
anova_model_data4 <- lm(income ~ degree * marital, data = hg)
anova_tway2 <- anova(anova_model_data4)
print(anova_tway2)
```

Analysis of Variance Table

```
Response: income
          Df      Sum Sq   Mean Sq F value    Pr(>F)
degree      4 2.1526e+11  5.3815e+10  56.7944 < 2.2e-16 ***
marital      4 4.4782e+10  1.1195e+10  11.8152 2.247e-09 ***
degree:marital 16 1.7599e+10  1.1000e+09   1.1609   0.2938
Residuals    969 9.1817e+11  9.4755e+08
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- degree ~marital, no significance income~degree * marital, significance interaction exists

non - parametric sets

wilcoxon rank_sum test(mann-whitney u test): compare medians of 2 independent

```
group00 <- c(5:7)
group000 <- c(8:10)
wilcon <- wilcox.test(group00, group000)
print(wilcon)
```

Wilcoxon rank sum exact test

```
data: group00 and group000
W = 0, p-value = 0.1
alternative hypothesis: true location shift is not equal to 0
```

- since the p-value>5%, we fail to reject null analysis and conclude that there is no difference significant between median of group (00 and 000)

kruskal-wallis test: compare median of more than one group

H0: median value of A=B=C

H1: median value of A!=B!=C

```
data222 <- data.frame(group6 = factor(c("A", "A", "A", "B", "B", "B", "C", "C", "C")),
  value = c(5:13))
head(data222)
```

```
group6 value
1      A      5
2      A      6
3      A      7
4      B      8
5      B      9
6      B     10
```

kruskal-wallis test

```
kruskal_results <- kruskal.test(value ~ group6, data = data222)
print(kruskal_results)
```

Kruskal-Wallis rank sum test

data: value by group6
Kruskal-Wallis chi-squared = 7.2, df = 2, p-value = 0.02732

- since p-value < 5%, we fail to reject the null hypothesis and conclude that there is a significant difference in median between group A, B and C

using gsssubset

```
kruskal_result2 <- kruskal.test(hg$income ~ hg$marital, data = data222)
print(kruskal_result2)
```

Kruskal-Wallis rank sum test

data: hg\$income by hg\$marital
Kruskal-Wallis chi-squared = 66.454, df = 4, p-value = 1.271e-13

- p-value < 5%, we fail to reject,